SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. A positronium atom is a system that consist of a positron and an electron that orbit each other. Choose the wrong statement.

(A) The Rydberg constant for positronium is half as large as it is for ordinary hydrogen atom.

(B) Wavelength in the positronium spectral lines are all twice of those corresponding in the hydrogen spectrum

(C) In positronium electron and positron rotate about common centre of mass

(D) Wavelengths in the positronium spectral lines are all half of those corresponding in the hydrogen spectrum.

2. Lyman alpha, the n=1 to n=2 transition in atomic hydrogen occurs at 1215 Å.

(A) Radiation of wavelength longer than 911 Å can photo–ionize hydrogen atom in ground state.

(B) Radiation of wavelength longer than 3645 Å can photo–ionize hydrogen atom in first excited state.

(C) Radiation of wavelength longer than 228 Å can photo–ionize He^+ ion in ground state.

(D) Radiation of wavelength longer than 1215 Å can photo–ionize He^+ ion atom in first excited state.

3. The maximum kinetic energy of photo-electron liberated from the surface of lithium (work function = 2.35 eV) by electromagnetic radiation whose electric component varies with time as $E = a[1 + \cos(2\pi f_1 t)]\cos(2\pi f_2 t)$ where 'a' is a constant, $f_1 = 3.6 \times 10^{15}$ Hz and $f_2 = 1.2 \times 10^{15}$ Hz is

[Take : $h = 6.6 \times 10^{-34}$ J-s]

(A) 2. 6 eV (B) 7.55 eV (C) 12.5 eV (D) 17.45 eV

4. Electrons with de-Broglie wavelength λ fall on the target in an X-ray tube. The cut-off wavelength of the emitted X-rays is **[JEE 2007]**

(A) $\lambda_0 = \frac{2mc\lambda^2}{h}$ (B) $\lambda_0 = \frac{2h}{mc}$

(C) $\lambda_0 = \frac{2m^2c^2\lambda^3}{h^2}$ (D) $\lambda_0 = \lambda$

5. A stationary hydrogen atom of mass M emits a photon corresponding to the longest wavelength of Balmer series. The recoil velocity acquired by the atom is
(R = Rydberg constant and h = plank's constant)

(A) $\frac{Rh}{M}$ (B) $\frac{Rh}{4M}$ (C) $\frac{3}{4}\frac{Rh}{M}$ (D) $\frac{5}{36}\frac{Rh}{M}$

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6. In an X-ray experiment target is made up of copper (Z = 29) having some impurity. The K_α line of copper have wavelength λ_0. It was observed that another K_α line due to impurity have wavelength $\frac{784}{625}\lambda_0$. The atomic number of the impurity element is

(A) 22 (B) 23 (C) 24 (D) 26

7. Which one of the following statements is **WRONG** in the context of X-rays generated from a X-ray tube ? **[JEE 2008]**

(A) Wavelength of characteristic X-rays decreases when the atomic number of the target increases

(B) Cut-off wavelength of the continuous X-rays depends on the atomic number of the target

(C) Intensity of the characteristic X-rays depends on the electrical power given to the X-rays tube

(D) Cut-off wavelength of the continuous X-rays depends on the energy of the electrons in the X-ray tube

8. A parallel beam of photons of intensity I strike a plane surface at angle 45° to the surface. If plane surface have reflection coefficient 0.5 then pressure exerted by the beam on the surface (C is speed of light) is :

(A) $\frac{3I}{4C}$ (B) $\frac{3I}{2C}$ (C) $\frac{I}{2\sqrt{2}C}$ (D) $\frac{3I}{2\sqrt{2}C}$

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9. A photoelectric material having work-function ϕ_0 is illuminated with light of wavelength $\lambda \left(\lambda < \frac{hc}{\phi_0}\right)$.

The fastest photoelectron has a de-Broglie wavelength λ_d. A change in wavelength of the incident light by $\Delta\lambda$ results in a change $\Delta\lambda_d$ in λ_d. Then the ratio $\Delta\lambda_d/\Delta\lambda$ is proportional to **[JEE Advanced-2017]**

(A) λ_d^3/λ^2 (B) λ_d^3/λ (C) λ_d^2/λ^2 (D) λ_d/λ

Multiple Correct Answer Type 7 Q. [4 M (–1)]

10. Photons of wavelength 248 nm fall on a metal surface whose work function is 2.2 eV. Assume that each photoelectron inside the metal lattice may come out of the surface or collide with the lattice before coming out. In each collision with the lattice, it loses 20% of its existing energy. Which of the following can be a kinetic energy of an ejected photoelectron?

(A) 2.8 eV (B) 2.24 eV (C) 1.8 eV (D) 1 eV

11. Applying Bohr's model choose the correct statement(s) :-

(A) K_β photon of aluminium will be more energetic than K_α photon of Lithium

(B) K_α photon of Beryllium will be more energetic than K_α photon of Lithium

(C) K_α photon of sodium will be more energetic than K_α photon of magnesium

(D) K_α photon of aluminium will be less energetic than K_β photon of aluminium

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12. A moving neutron collides with stationary H-atom in ground state. As a result it excites and then de excites. The corresponding radiation fall on a surface, having work function σ. The minimum value of required kinetic energy for neutron is E_0 and possible minimum value of de broglie wavelength of emitted photoelectrons is λ_0. If neutron hits stationary He^+ ion instead of stationary H atom, then minimum value of kinetic energy for neutron is E_1. Then

(A) The value of energy transferred from neutron to H-atom is $\frac{3E_0}{4}$.

(B) The value of λ_0 is $\frac{h}{\sqrt{m_e(E_0 - 2\sigma)}}$.

(C) The value of E_1 is $4E_0$

(D) The value of E_1 is $\frac{5E_0}{2}$

13. A particle of mass 6.6×10^{-30} kg starts (t = 0) moving on a straight line with velocity 10 m/s. Its velocity decreases with time, however rate of change of de-Broglie wavlength associated with particle remains constant at 10^{-4} m/s. (Take h = 6.6×10^{-34} J–s) :-

(A) Velocity of particle at t = 0.9s is 1 m/s

(B) Velocity of particle at t = 0.9s is 6 m/s

(C) Magnitude of retaration of particle at t = 0.9 s is 4 m/s^2

(D) Magnitude of retaration of particle at t = 0.9 s is 1 m/s^2

14. Whenever a photon is emitted by a hydrogen atom in Balmer series

(A) it is always followed by another photon in Lyman series

(B) another photon emitted will always have wavelength of approximately1225Å.

(C) it may emit a second photon, but no further information can be obtained about it.

(D) it will not emit another photon.

15. The potential difference applied to an X-ray tube is increased. As a result, in the emitted radiation:

(A) the intensity increases

(B) the minimum wave length increases

(C) the intensity decreases

(D) the minimum wave length decreases

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16. Highly excited states for hydrogen like atoms (also called Rydberg states) with nuclear charge Ze are defined by their principal quantum number n, where n >> 1. Which of the following statement(s) is (are) true? **[JEE Advanced-2016]**

(A) Relative change in the radii of two consecutive orbitals does not depend on Z

(B) Relative change in the radii of two consecutive oribitals varies as 1/n

(C) Relative change in the energy of two consecutive orbitals varies as $1/n^3$

(D) Relative change in the angular momenta of two consecutive orbitals varies as 1/n

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]

(Single Correct Answer Type)

Paragraph for Question 17 to 19

A mercury arc lamp provides 0.1 watt of ultra-violet radiation at a wavelength of $\lambda = 2537$ Å only. The photo tube (cathode of photo electric device) consists of potassium and has an effective area of 4 cm^2. The cathode is located at a distance of 1m from the radiation source. The work function for potassium is $\phi_0 = 2.22$ eV.

17. According to classical theory, the radiation from arc lamp spreads out uniformly in space as spherical wave. What time of exposure to the radiation should be required for a potassium atom (radius 2Å) in the cathode to accumulate sufficient energy to eject a photo-electron ?

(A) 352 second (B) 176 second (C) 704 seconds (D) No time lag

18. To what saturation current does the flux of photons at the cathode corresponds if the photo conversion efficiency is 5%.

(A) 32.5 nA (B) 10.15 nA (C) 65 nA (D) 3.25 nA

19. What is the cut off potential V_0 ?

(A) 26.9 V (B) 2.69 V (C) 1.35 V (D) 5.33 V

Paragraph for Questions no. 20 to 22

A light of wavelength λ is incident on a metal sheet of work function $\phi = 2$ eV. The wavelength λ varies with time as $\lambda = 3000 + 40t$, where λ is in Å and t is in second. The power incident on metal sheet is constant at 100 W. This signal is switched on and off for time intervals of 2 minutes and 1 minute respectively. Each time the signal is switched on, the λ start from initial value of 3000 Å. The metal plate is grounded and electron clouding is negligible. The efficiency of photoemission is 1%

(hc = 12400 eVÅ, 1eV = 1.6×10^{-19} J)

20. The time after which photoemission will stop is :-

(A) 120 s (B) 80 s (C) 60 s (D) 180 s

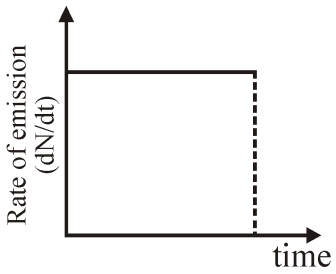
21. The total number of photoelectrons ejected in one hour is :-

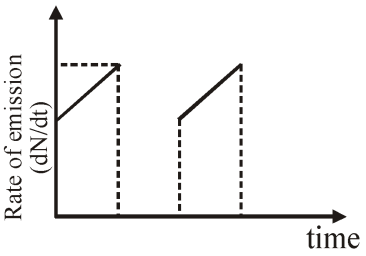
(A) 3.71×10^{17} (B) 3.71×10^{21} (C) 3.71×10^{25} (D) 3.71×10^{7}

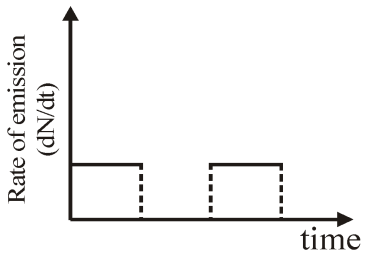
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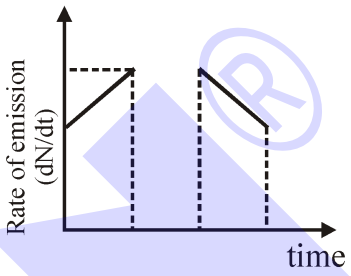
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22. The variation of rate of emission vs time is :-

(A) 

(B) 

(C) 

(D) 

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 5 Q. [4 M (0)]

23. An electron is confined to a tube of length L. Its potential energy in first half of the tube is zero, while the potential energy in other half is 12 eV. If the total energy of electron is 16 eV, find the ratio of the longest to the shortest de Broglie wavelength of an electron in the tube.

24. A potential difference of V volts is applied on two parallel electrodes separated by a distance of 4.0×10^{-2} m . The electrons of very low energy are injected into the region between the electrodes which contains argon at low pressure . The average distance the electrons travel between collisions with argon atoms is 8×10^{-5} m . The ionization energy of argon atom is 16 eV . Estimate the minimum value of V(in kV) such that the electrons will cause ionization in argon atoms by collision .

25. When 0.50 Å X-rays strike a material, the photoelectrons from the K shell are observed to move in a circle of radius 23 mm in a magnetic field of 2×10^{-2} T. The binding energy of K-shell electrons is $a^2 \times 100$ eV then write the value of a in your OMR sheet ?

(Take : mass of electron = 9.2×10^{-31} kg & hc = 12400 eV–Å)

26. Consider a universe in which the π-meson orbits around the nucleus instead of electron. Assuming a Bohr model for a π-meson of mass m_π and of the same charge as the electron is in a circular orbit of radius r about the nucleus with an orbital angular momentum $\frac{h}{2\pi}$. If the radius of a nucleus of atomic number Z is given by $R = 1.6 \times 10^{-15} Z^{1/3}$ m. The total number of elements in this universe that can exist bis given as 'N'. Fill $\left[\frac{N-1}{12}\right]$ in OMR sheet. [Given $\frac{\varepsilon_0 h^2}{\pi m_e e^2} = 0.53\text{Å}$; $\frac{m_\pi}{m_e} = 265$; neglect any shielding effect for the havier atoms and assume non relativistic physics to be applicable and take $5^{1/4} \approx 1.5$]

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27. A silver sphere of radius 1 cm and work function 4.7 eV is suspended from an insulating thread in free-space. It is under continuous illumination of 200 nm wavelength light. As photoelectrons are emitted, the sphere gets charged and acquires a potential. The maximum number of photoelectrons emitted from the sphere is $A \times 10^z$ (where $1 < A < 10$). The value of 'Z' is **[JEE 2011]**

Subjective Type

2 Q. [4 M (0)]

28. A beam of light has three wavelengths 4144Å, 4972Å & 6216 Å with a total intensity of 3.6×10^{-3} W.m^{-2} equally distributed amongst the three wavelengths. The beam falls normally on an area 1.0 cm^2 of a clean metallic surface of work function 2.3 eV. Assume that there is no loss of light by reflection and that each energetically capable photon ejects one electron. Calculate the number of photoelectrons liberated in two seconds.

29. Assume that the de-Broglie wave associated with an electron can form a standing wave between the atoms arranged in a one dimensional array with nodes at each of the atomic sites. It is found that one such standing wave is formed if the distance 'd' between the atoms of the array is 2 Å. A similar standing wave is again formed if 'd' is increased to 2.5 Å but not for any intermediate value of d. Find the energy of the electrons in electron volts and the least value of d for which the standing wave of the type described above can form.

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GR # MODERN PHYSICS-1 **ANSWER KEY**

SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (D) 2. Ans. (A) 3. Ans. (D) 4. Ans. (A) 5. Ans. (D) 6. Ans. (D)

7. Ans. (B) 8. Ans. (A) 9. Ans. (A)

Multiple Correct Answer Type **7 Q. [4 M (–1)]**

10. Ans. (A, C, D) 11. Ans. (A,B,D) 12. Ans. (A,B,D) 13. Ans. (A, D)

14. Ans. (A,B) 15. Ans. (A,D) 16. Ans. (A, B, D)

Linked Comprehension Type **(2 Para × 3Q.)** **[3 M (-1)]**

(Single Correct Answer Type)

17. Ans. (A) 18. Ans. (A) 19. Ans. (B) 20. Ans. (B) 21. Ans. (B) 22. Ans. (B)

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

23. Ans. 2 24. Ans. 8 25. Ans. 8 26. Ans. 3 27. Ans. 7

Subjective Type **2 Q. [4 M (0)]**

28. Ans. 1.1×10^{12} 29. Ans. KE $\cong$ 151 eV, d_{least} = 0.5 Å

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